

Excessive Progression in Weekly Running Distance and Risk of Running-related Injuries: An Association Modified by Type of Injury

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32 **Study Design:** An explorative 1-year prospective cohort study.

33 **Objective:** To examine whether an association between a sudden change in weekly running
34 distance and running-related injury is modified by type of injury.

35 **Background:** It is widely accepted that a sudden increase in running distance is strongly related to
36 injury in runners. But the scientific knowledge supporting this assumption is limited.

37 **Methods:** A volunteer sample of 874 healthy novice runners, who started a self-structured running
38 regimen, were provided a global positioning system (GPS) watch. After each running session
39 during the study period, participants were categorized into 1 of the following exposure groups based
40 on their progression in weekly running distance: less than 10% (including regression), 10 to 30%, or
41 more than 30%. The primary outcome was running-related injury.

42 **Results** A total of 202 runners sustained a running-related injury. Using Cox regression analysis, no
43 statistically significant differences in injury rates were found across the 3 exposure groups. An
44 increased rate of distance-related injuries (patellofemoral pain, iliotibial band syndrome, medial
45 tibial stress syndrome, gluteus medius injury, greater trochanteric bursitis, injury to the tensor fascia
46 latae, and patellar tendinopathy) existed in those progressing their weekly running distance by more
47 than 30% compared with those progressing less than 10% (hazard ratio [HR] = 1.59 [95% CI: 0.96
48 to 2.66], $P = 0.07$).

49 **Conclusion:** Novice runners progressing their running distance by more than 30% over a 2-week
50 period seem to be more vulnerable to distance-related injuries than runners increasing their running
51 distance by less than 10%. Owing to the exploratory nature of the present study, randomized
52 controlled trials are needed to verify these results, and more experimental studies are needed to
53 validate the assumptions. Still, novice runners may be well advised to progress their weekly
54 distances by less than 30% per week over a 2-week period.

55 **Keywords:** etiology, novice, prospective

Introduction

Running-related injuries are common, especially in novice runners.^{4, 6, 23} From a preventive perspective, knowledge about the underlying mechanisms leading to such injuries is needed,³³ as there is currently very limited knowledge about injury prevention.³⁴ Interestingly, Hreljac¹⁵ pragmatically stated that all overuse injuries in runners are linked to training errors. This statement is supported by others who work with causality and injury in sport.^{19, 30}

Training error is, unfortunately, a vague term because it covers a broad range of flaws in the running regimen. To name a few, these flaws may include excessive mileage, a rapid change in intensity, and a sudden increase in running distance. While difficult to define, it is a general belief that a sudden excessive increase in running distance is one that overwhelms the ability for adaptive changes and tissue repair which ultimately leads to an injury.¹⁵ To date, the scientific knowledge to support a link between sudden increases in weekly mileage and injury development is extremely limited.²³ Yet, Jacobs et al¹⁶ reported that one third of runners with injuries described that they had changed their running routines just prior to their injury development.

The so-called 10% rule is commonly used as a guideline for a maximum training progression by runners, coaches, and clinicians.¹⁷ Although the 10% rule is widely accepted, a well-designed randomized controlled trial failed to identify an increased risk of injury in novice runners progressing their weekly running distance by 24% over a 8-week period compared to those progressing their weekly running distance by the recommended 10% over a 12-week period.⁶ Based on the results of this study, the 10% rule and its relationship to injury development should be reconsidered. Maybe, cumulative progressions over 8 and 12 weeks are not sudden enough. Possibly, changes in running routines over for instance a 2-week period may more likely lead to injuries. In addition, a higher progression rate may be needed to identify a difference in injury risk. In a recent case-control analysis, injured runners had an average progression in weekly distance prior to injury of 31% whereas the non-injured controls had a progression of 22%.²⁴ Unfortunately, the sample size in the study was small ($n = 60$) and only 13 runners sustained an injury. It was, therefore, concluded that the results from that study should be interpreted with extreme caution, and the need for more large-scale prospective studies was emphasized.

Also, including all types of running-related injuries into a statistical analysis may be problematic. Different running-related injuries may develop based on the type of errors in the running regime. It seems plausible that some injuries may more likely develop owing to sudden increases in running distance, while others may more likely develop owing to a sudden increase in running pace.²⁵ A biomechanical rationale for the possible underlying mechanism for the development of some pace-related injuries has been suggested³ and a number of studies have directly linked fast pace or sprinting to the following injuries: Achilles tendinopathy,¹⁴ hamstring strains,^{8,13} tibial stress fractures,^{10,11} and iliopsoas strains.¹

Conversely, sudden increases in running distance may be linked with other types of running-related injuries: iliotibial band syndrome,²⁰ patellofemoral pain,³² patellar tendinopathy,¹² and medial tibial stress syndrome.¹⁸ Biomechanically, a recent study by Schache et al²⁹ can be used to provide a rationale behind the mechanisms leading to injuries in the anterior part of the knee: During slow-speed running, the cumulative load at the knee joint is higher than the cumulative load during faster running. If runners run slower at longer distances (especially when fatigued), this prompts the hypothesis that a sudden increase in mileage is associated with an increased risk of running-related injuries in the anterior part of the knee. Possibly, similar assumptions could be made for other anatomical locations like the lateral part of the hip and the medial part of the lower leg. Accordingly, distance-related injuries could be hypothesized to be patellofemoral pain, iliotibial band syndrome, medial tibial stress syndrome, gluteus medius injury, greater trochanteric bursitis, injury to the tensor fascia latae, and patellar tendinopathy.

To determine if a sudden increase in running distance is associated with the development of running-related injuries, one could, for instance, follow a large cohort of runners prospectively, register their running routines using a valid method,²⁴ calculate the change in weekly running distance over a 2-week period, diagnose each injury by clinical examination, and divide the injuries into categories prior to analysis. Using this approach, it may be possible to determine whether the hazard of sustaining specific distance-related injuries is increased in novice runners progressing their weekly running distance by more than 30% just prior to the occurrence of injury. The purpose of the present study was, therefore, to examine whether an association between a sudden change in weekly running distance and running-related injury is modified by type of injury. We hypothesized that the injury rate would be significantly higher in novice runners who had a progression of weekly

running distance of more than 30%, than in novice runners with a progression of less than 10% and that the increased injury rate would only be present for distance-related injuries like patellofemoral pain, iliotibial band syndrome, medial tibial stress syndrome, gluteus medius injury, greater trochanteric bursitis, injury to the tensor fascia latae, and patellar tendinopathy.

METHODS

The Danish Novice Running project (DANO-RUN) was a prospective, observational cohort study with a 1-year follow-up.²⁶ The study protocol was presented to the local ethics committee Central Denmark Region, who waived the request for approval because observational studies do not need an ethics approval according to Danish laws. Recruitment procedures and inclusion and exclusion criteria have been presented in previously published work.^{21, 22} Of the 1530 individuals who volunteered for the study, 579 were excluded prior to baseline evaluation. At baseline investigation, an additional 18 individuals were excluded due to a variety of reasons. A detailed description of the reasons for excluding the 597 individuals has been presented elsewhere.²² Finally, 933 persons (464 females, 469 males) were enrolled in the study after signing an informed consent.

Baseline investigation

At baseline, body mass index (BMI) was calculated based on height measured with a ruler and weight measured with a calibrated scale (SC 330; Tanita Corporation, Tokyo, Japan). Subsequently, the participants were provided with a pair of neutral running shoes (Supernova Glide 3 Male or Female, Adidas, Herzogenaurach, Germany) and were asked to use this shoe in all running sessions during the following year. In addition, each participant was provided with a global positioning system (GPS) watch (Forerunner 110 M, Garmin International Inc., Olathe, Kansas, USA), which has previously been shown to accurately measure the distance covered by runners.²⁴ Participants were instructed to upload every training session (running only) they completed during the year to an internet-based training diary (<http://www.vilober.dk/>). In case of missing GPS data, they were told to upload the time and distance manually. After the baseline evaluation, the participants initiated their running program which was self-structured. The participants decided for themselves when and where to run, with no restrictions with regard to time, duration, and intensity of each training session. Similarly, participants decided for themselves the change in weekly running distance.

150 Participants were told that they would receive the shoes and GPS for free if they completed a
151 minimum of 52 running sessions during the 1-year follow-up.

153 **Running-related injuries**

154 The primary outcome was the first running-related injury to occur during the 1 year period. A
155 running-related injury was defined as “Any musculoskeletal complaint of the lower extremity or
156 back caused by running that restricted the amount of running (distance, duration, pace, or
157 frequency) for at least 1 week.” This definition was a modified version of the definition used by
158 Buist et al.⁶ If the participants sustained a running-related injury during the study period, they were
159 instructed to contact the medical team via their personal training diary. Then, the participant was
160 contacted by telephone, and an appointment for a clinical examination was made during which
161 he/she was examined by a physiotherapist, preferably no later than 1 week after the telephone
162 conversation. A standardized examination procedure and non-validated guidelines for the diagnostic
163 criteria were used to classify each injury. If the physiotherapist was unable to diagnose the injury at
164 the clinical examination, an additional examination including diagnostic imaging was performed at
165 a nearby hospital, which was needed for approximately 25% of all injured cases.

166
167 At each clinical examination, the running-related injury was classified as being based on overuse or
168 on trauma. Injuries occurring during the first 2 weeks of the running program were excluded (n =
169 34) because 1) we hypothesized that these injuries probably occurred because the initial running
170 distance during the first week was too long and 2) it was not possible to calculate the progression in
171 running distance during the first 2 weeks. A total of 873 of the 933 individuals originally included
172 in the DANO-RUN study were included in the present analyses after excluding those sustaining
173 injuries in the first 2 weeks and after excluding 23 uninjured individuals who left the study for other
174 reasons than injury in the first two weeks. A flow chart of the exclusions prior to the analyses is
175 provided in **FIGURE 1**.

176
177 The injured participant was asked to report the specific day or training session at which the
178 symptoms started. This day was identified as the day the injury occurred. Patellofemoral pain,
179 iliotibial band syndrome, and patellar tendinopathy have previously been linked to excessive
180 running distance.²⁵ Additionally, we hypothesized the development of medial tibial stress syndrome,
181 gluteus medius injury, greater trochanteric bursitis, and injury to the tensor fascia latae to be linked

182 to excessive progression in running distance. The assumption that these injuries are likely to be
183 distance-related and not pace-related remains to be further validated, and as a consequence, the
184 analyses presented in this paper should be considered as explorative.

185

186 **Change in weekly running distance**

187 The primary exposure of interest was the ratio between 2 weekly distances given as a percentage of
188 change. The ratio was calculated in the following manner. After each running session during the
189 study period, the sum of km from a running session was added to the sum of km covered in a 6-day
190 period prior to that session. Accordingly, the cumulative number of km over a 1-week period
191 (named week 1) was determined. Then, the cumulative running distance from day 7 to 13 prior to
192 the training session of interest was calculated. Based on these 2 absolute distances from week 1 and
193 week 0, the progression or regression (in %) between these 2 periods was calculated by dividing the
194 2 absolute distances and then multiplying the results by 100. Ratio between weekly distances =
195 (Total running distance week 1) / (Total running distance week 0) * 100.

196

197 This approach makes the ratio between weekly distances a covariate that is time-dependent in the
198 sense that progression (a positive ratio) or regression (a negative ratio) will change many times
199 during study period, effectively after each running session. This enables data analysis of a time-
200 dependent exposure variable that allows the participant to change/switch/transit between exposure
201 groups (or stay in the same exposure group) each time they run. Importantly, this approach is much
202 different than the average change in running distance (which is not time-dependent) that has been
203 used as the exposure of interest in many of the previously published studies.²³

204

205 After calculating the ratio after each running session, the change in weekly distance was categorized
206 into 1 of the following 3 exposure groups (in statistical terms: exposure states): progression (or
207 regression) less than 10%, progression between 10% and 30%, and progression greater than 30%.
208 The 10% cut-off was chosen based on the general belief that a graded training program may become
209 injurious at a progression in weekly distance of more than 10%.²³ The 30% cut-off was chosen
210 based on the findings from a pilot study.²⁴ Because most participants varied in their running routine,
211 each runner was able to switch/move between the 3 exposure groups (less than 10%, 10 to 30%, or
212 greater than 30%) every time they completed a new running session during the study period.
213 Statistically, such switches between exposure groups are known as multi-state transitions.²⁸

214 Importantly, when a hazard ratio (HR) was calculated in the present study, it was based solely on
215 the 3 exposure groups with the goal to answer the following question: is it hazardous to progress by
216 more than 30% over a 2 week period?

217
218 If the runner did not run in week 0, it was not possible to calculate a ratio between the weekly
219 distances in week 1 and week 0 because the denominator was zero. Then, participants were
220 categorized into a “progression or regression not available” (NA) group. To summarize, participants
221 were continuously (after each running session) able to be categorized into 1 of 4 groups:
222 progression or regression from week 1 to week 0 of less than 10% (reference group); progression
223 from week 1 to week 0 between 10 to 30%; progression from week 1 to week 0 of greater than
224 30%; or progression not possible to calculate (NA)

225

226 **Statistical analysis**

227 The HRs between exposure groups and running-related injury were estimated using Cox regression,
228 with the number of km during the training sessions as the time scale. In statistics, an exposure that
229 can change over time is known as a “time-dependent covariate”. This concept enables each
230 participant to switch between exposure states continuously (after each running session) using
231 delayed entry in the Cox regression model.²⁸ In the analyses, cause-specific hazards of the
232 instantaneous risk of injury from a specific injury category (distance-related injuries, pace-related
233 injuries, other overuse injuries, and traumatic injuries) were calculated using competing risks. The
234 HRs were compared across injury categories with the less than 10% group as the reference group.
235 The unit of analysis was each runner. Participants were censored in case of pregnancy, disease, lack
236 of motivation, unwillingness to attend clinical examination for running-related injury, or end of
237 follow-up at 1 year, whichever came first. The assumption of proportional hazards was evaluated by
238 log-minus-log plots. In addition, at least 10 injuries per predictor variable (EPV) included in the
239 Cox regression analysis must be present to perform a valid statistical analysis.²⁷ Because a
240 considerable number of the events were used to perform the crude analysis, adjusted analysis that
241 included more confounders than previous running-related injury, BMI, fat percentage, and age
242 could not be performed. Because the hazards were not proportional in the NA exposure group, the
243 data from this exposure group are not presented in the main results. HRs were considered
244 statistically significant when $P \leq 0.05$. In addition, for proper interpretation of study results,

estimated effect size and estimated precision (95% confidence limits) were calculated.³¹ All analyses were performed using STATA version 12.

RESULTS

A total of 202 of the 873 novice runners included in the study sustained a running-related injury. Demographic characteristics of the runners stratified by injury status are presented in **TABLE 1**. During the study period, the participants ran a total of 148 491 km in 35 410 training sessions. After completing each running session, it was possible to calculate which exposure group the participant had at that given time point: for 16 253 sessions (46%) the progression (or regression) was less than 10%; for 1939 sessions (5.5%) the progression was between 10% and 30%; and for 7507 sessions (21%) the progression was greater than 30%. Finally, it was not possible to calculate a progression for 9711 running sessions (27.5%).

The primary result revealed that the injury rate was modified by injury diagnoses in the greater than 30% progression exposure group based on the results from the analysis of pace-related injuries (**TABLE 2**, analysis C) and distance-related injuries (**TABLE 2**, analysis B) being significantly different ($P < 0.05$). In the analysis including all injuries, no significant differences in injury rates were found across exposure groups (**TABLE 2**, analysis A). However, if injuries were restricted to patellofemoral pain, iliotibial band syndrome, medial tibial stress syndrome, patellar tendinopathy, gluteus medius injury, greater trochanteric bursitis, and injury to the tensor fascia latae (**TABLE 2**, analysis B), an increased rate of sustaining an injury was observed in those runners progressing their weekly running distance by greater than 30% compared with those progressing less than 10% (HR = 1.59 [95% CI: 0.96 to 2.66], $P = 0.07$). The estimate minimally changed after adjusting for age, BMI, fat percentage, and previous history of running-related injury (HR = 1.59 [95% CI: 0.96 to 2.68], $P = 0.07$). Although non-significant, the estimates from analysis including plantar fasciitis, Achilles tendinopathy, tibial stress fracture, hamstring injuries, iliopsoas injuries, and gastrocnemius and soleus injuries revealed that the injury rate decreased in relation to increasing progression in running distance (**TABLE 2**, analysis C). A plot of the cause-specific injury-probability is presented in **FIGURE 2**. The observed and expected numbers of injuries in each exposure group are presented in **TABLE 3**.

DISCUSSION

The present study revealed that novice runners progressing their weekly running distance by greater than 30% were more vulnerable to distance-related injuries like patellofemoral pain, iliotibial band syndrome, medial tibial stress syndrome, patellar tendinopathy, gluteus medius injury, greater trochanteric bursitis, and injury to the tensor fascia latae, than runners whose progression in running distance was less than 10% (TABLE 2, analysis B). Such a relationship was not present for other injury types like traumatic injuries (TABLE 2, analysis E) or pace-related injuries like Achilles tendinopathy, plantar fasciitis, tibial stress fracture, hamstring injuries, iliopsoas injuries, and calf injuries (TABLE 2, analysis C). Interestingly, no dose-response relationship was revealed because it was equally hazardous to increase the weekly running distance by 10 to 30% compared to less than 10%. Accordingly, a greater difference in progression rates than 10% versus 10 to 30% may, therefore, be needed to adequately evaluate the hazard involved in a sudden increase in weekly running distances over a 2-week period. In addition, only specific injuries must be included in the analysis to reveal any differences in hazards between exposure groups.

Clinically, this knowledge must be used with caution when designing a running program because multiple switches (in statistical terms: multistate transitions²⁸) over time may affect injury risk considerably. In the present study, we examined 1 switch over a 2-week period (from week 0 to week 1) in relation to injury risk. We did not take into account the cumulative distance in the week prior (labeled “week -1”) and the change in progression/regression from “week -1” to “week 0” or the change in progression from “week -2” to “week -1”. In the present study, the 30% cut-off was found to be associated with an increased injury rate. If, for example, a runner increased his/her weekly distance by 36%, an increased injury rate existed for “distance-related injuries”. If the increase remained less than 30%, the injury rate was less, which could lead the runner to assume that the injury rate would be low no matter how he/she scheduled his/her running if he/she just kept the increase to less than 30%. Such an assumption must, however, be drawn with extreme caution. The injury rate may be low when increasing the weekly distance by for instance 5%, then by 7%, and finally by 3%. But the injury rate may be high by progressing from 25%, to 25%, and finally to 25% over a 4-week period, even though the progression did not exceed 30%. Clinicians should, therefore, be cautious about advising runners to increase their running distance at a high progression level even though our results suggest the runner to be at low risk if the progression does not exceed 30%. For now, it seems plausible to assume that continuous switches over time between high

308 progression states (for example 25%) may ultimately lead to an injury. But more work is needed to
309 validate this assumption.

310

311 The biomechanical rationale that explains the mechanisms leading to injury development owing to a
312 sudden increase in weekly distance remains unknown. Yet, it seem plausible to assume that running
313 longer distances than usual may force the runner to decrease the running speed especially if they are
314 fatigued. If running speed is decreased for a given running distance, the cumulative number of steps
315 increase considerably.⁹ This is important, because the cumulative load at the knee joint seems to be
316 significantly increased during slow speed running compared with faster running.²⁹ A similar finding
317 may be present at the lateral part of the hip and at the medial part of the lower leg. In contrast,
318 another biomechanical rationale suggests a sudden increase in running speed to be linked with
319 injuries like Achilles tendinopathy, plantar fasciitis, tibial stress fracture, hamstring injuries,
320 iliopsoas injuries, and calf injuries.^{3, 9, 29} This may be the reason for the difference in the results
321 across distance-related and pace-related injuries. We strongly underline the importance of this
322 assumption because it seems crucial to make such a categorisation if the influence of training errors
323 on injury development should be investigated properly.

324

325 The strengths of the present study are the number of individuals included and the method used to
326 gather information about running exposure. Also, the statistical approach used allowing for delayed
327 entry / time-dependent exposures and competing risks is a major strength. A comprehensive tutorial
328 of the strengths and validity of time-to-event analysis including a description of time-dependent
329 exposures in observational studies is provided by Bull and Spiegelhalter.⁷ But important limitations
330 should be taken into account when interpreting the results. First, nearly all novice runners included
331 in the study utilized a rear-foot strike.² The results from the present study cannot be generalized to
332 runners utilizing other foot strike patterns, as other mechanisms leading to injury may exist for
333 those with a midfoot or forefoot strike pattern. Second, it must be emphasized that the present study
334 is an observational study. Because of this study design, the results presented are vulnerable to
335 confounding, and the results must, therefore, be interpreted with caution. In **TABLE 1**, age, BMI,
336 fat percentage, and previous running-related injuries were found to be risk factors for developing an
337 injury. These risk factors were included in an adjusted analysis, but the adjustment had little
338 influence on the estimates. Still, various unknown risk factors may confound the results presented in
339 the present study. To take into account the risk of confounding, more randomized controlled trials

are needed. In such studies, it is of great importance to perform clinical examination of the injured participants because different training approaches (sudden change in running distance versus sudden change in running pace) are associated with development of different types of injuries. Third, the results of the present study revealed that sudden progression of weekly running distance was a risk factor for the development of only 76 of 202 injuries (37.6%). Therefore, work has to be devoted to identify the training errors linked to the development of the remaining 126 injuries. For now, it seems plausible that excessive pace may be, somehow, associated with Achilles tendinopathy, plantar fasciitis, tibial stress fracture, hamstring injuries, iliopsoas injuries, and injuries to the triceps surae muscles ($n = 58$, 28.7%).²⁵ It would, therefore, be of great interest to perform an analysis to investigate whether the progression in running pace was associated with development of these injuries. Unfortunately, the measurement errors provided by the non-differential GPS watches used to quantify the running pace were too severe to perform such analysis based on the DANO-RUN dataset.

CONCLUSION

Novice runners increasing their weekly running distance by greater than 30% were more vulnerable to sustain distance-related injuries like patellofemoral pain, iliotibial band syndrome, medial tibial stress syndrome, patellar tendinopathy, gluteus medius injury, greater trochanteric bursitis, and injury to the tensor fascia latae than runners who kept their progressions in running distance to less than 10%. These differences were not present for what we would consider as “pace-related” or “other overuse” injuries. Due to the exploratory nature of the present study, randomized controlled trials are needed to confirm these findings, and more experimental studies are needed to validate the assumption that some structures are relatively more exposed to stress than other structures in relation to an increase in running distance. Yet, novice runners may be well advised to increase their weekly running distance by less than 30% over a 2-week period.

367 KEY POINTS

368

369 Findings

- 370 - A sudden increase in weekly running distance, of greater than 30%, over a 2-week period
371 may lead to the development of patellofemoral pain, iliotibial band syndrome, medial tibial
372 stress syndrome, patellar tendinopathy, gluteus medius injury, greater trochanteric bursitis,
373 and injury to the tensor fascia latae.

374

375 Implications

- 376 - Novice runners may be well advised to increase their weekly distance by less than 30%
377 over a 2 week period.
- 378 - Clinicians should pay particular attention to a runners' progression in weekly distance if
379 they have patellofemoral pain, iliotibial band syndrome, medial tibial stress syndrome,
380 patellar tendinopathy, gluteus medius injury, greater trochanteric bursitis, and injury to the
381 tensor fascia latae.

382

383 Caution

- 384 - Other training-related factors like progression over several weeks and intensity should be
385 taken into account when designing a program for runners.

386

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392 possible to buy GPS watches and running shoes at a reduced price.

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TABLE 1: Characteristics of the 873 participants stratified by injury status.

Variable	Unit	All (n = 873)	Injury free (n = 671)	Injured (n = 202)	P value
<i>Gender^a</i>	M / F	432 / 441	335 / 336	106 / 96	0.59
<i>Age^b (mean, sd)</i>	Years	37.2 ± 10.3	36.7 ± 10.2	39.0 ± 10.3	<0.01
<i>BMI^b (mean, sd)</i>	Kg / m ²	26.1 ± 4.2	25.9 ± 4.3	26.6 ± 4.2	0.05
<i>Fat pct^b (mean, sd)</i>	%	28.1 ± 9.3	27.6 ± 9.4	29.5 ± 9.3	0.01
<i>Previous RRI^a</i>	Yes / No	156 / 717	105 / 566	51 / 151	<0.01
<i>Previous non RRI^a</i>	Yes / No	327 / 546	242 / 429	85 / 117	0.13

Abbreviations: BMI, body mass index; Fat pct, fat percentage measured by bio-electrical impedance; RRI, running-related injury.

All dichotomious variables are presented as counts. The *P* values correspond to the tests for difference between injured and injury free. All variables were measured or reported prior to or at the baseline investigation. a = chi² test used to compare injured and injury free. b = un-paired t-test used to compare injured and injury free.

490 **Table 2:** Results from the Cox regressions linking injury rate with distance progression.

Exposure state	Hazard ratio	95% CI	P-value
A) All injuries (n=202)			
Less than 10%	1 (reference)		
10% to 30%	0.99	0.55 to 1.82	0.99
More than 30%	1.17	0.84 to 1.63	0.36
B) Distance-related injuries (n = 76)			
Less than 10%	1 (reference)		
10% to 30%	1.03	0.37 to 2.90	0.96
More than 30%	1.59	0.96 to 2.66	0.07
C) Pace-related injuries (n = 58)			
Less than 10%	1 (reference)		
10% to 30%	0.91	0.32 to 2.63	0.86
More than 30%	0.83	0.44 to 1.57	0.56
D) Other overuse injuries (n = 52)			
Less than 10%	1 (reference)		
10% to 30%	0.63	0.15 to 2.70	0.53
More than 30%	0.84	0.40 to 1.77	0.65
E) Traumatic injuries (n = 16)			
Less than 10%	1 (reference)		
10% to 30%	3.16	0.62 to 16.25	0.17
More than 30%	2.09	0.61 to 7.20	0.24

491

492 A) all 202 injured novice runners (all injuries) were included in the analyses

493 B) included the 76 runners sustaining distance-related injuries (patellofemoral pain, iliotibial band
494 syndrome, medial tibial stress syndrome, patellar tendinopathy, gluteus medius injury, greater
495 trochanteric bursitis, and injury to the tensor fascia latae),

496 C) included the 58 runners who sustained pace-related injuries (plantar fasciitis, Achilles
497 tendinopathy, tibial stress fracture, hamstring injuries, iliopsoas injuries, and injuries to the triceps
498 surae muscles)

499 D) included the 52 runners who sustained injuries of the medial meniscus, other stress fractures, and
500 other overuse injuries not included in B or C
501 E) included the 16 runners sustaining traumatic injuries like ankle inversions injuries.
502 Due to the low number of injuries in the 10% to 30% progression exposure group (n = 2 to 4,
503 **TABLE 3**), the hazard ratios between this group and the reference group should be interpreted with
504 caution.
505

TABLE 3: Observed and expected number of running-related injuries in each of the exposure groups.

	All injuries n = 202	Distance injuries n = 76	Pace injuries n = 58	Other overuse injuries n = 52	Traumatic injuries n = 16
Less than 10%	92 (85.6)	32 (33.9)	31 (26.8)	24 (24.5)	5 (8.0)
10% to 30%	12 (13.2)	4 (4.1)	4 (3.8)	2 (3.2)	2 (1.0)
More than 30%	54 (53.3)	26 (17.1)	13 (13.5)	10 (12.3)	5 (3.8)
NA	44 (53.0)	14 (20.9)	10 (13.8)	16 (12.0)	4 (3.2)
<i>P</i> value	0.36	0.06	0.60	0.49	0.45

Numbers outside parentheses are the observed number of injuries and the numbers in parentheses are the expected number of injuries. The *P* values correspond to the log-rank test for equality of survivor functions. The injuries in the distance, pace, other overuse, and traumatic injury groups are similar to those described in **TABLE 2**. NA = Not able to calculate a progression.

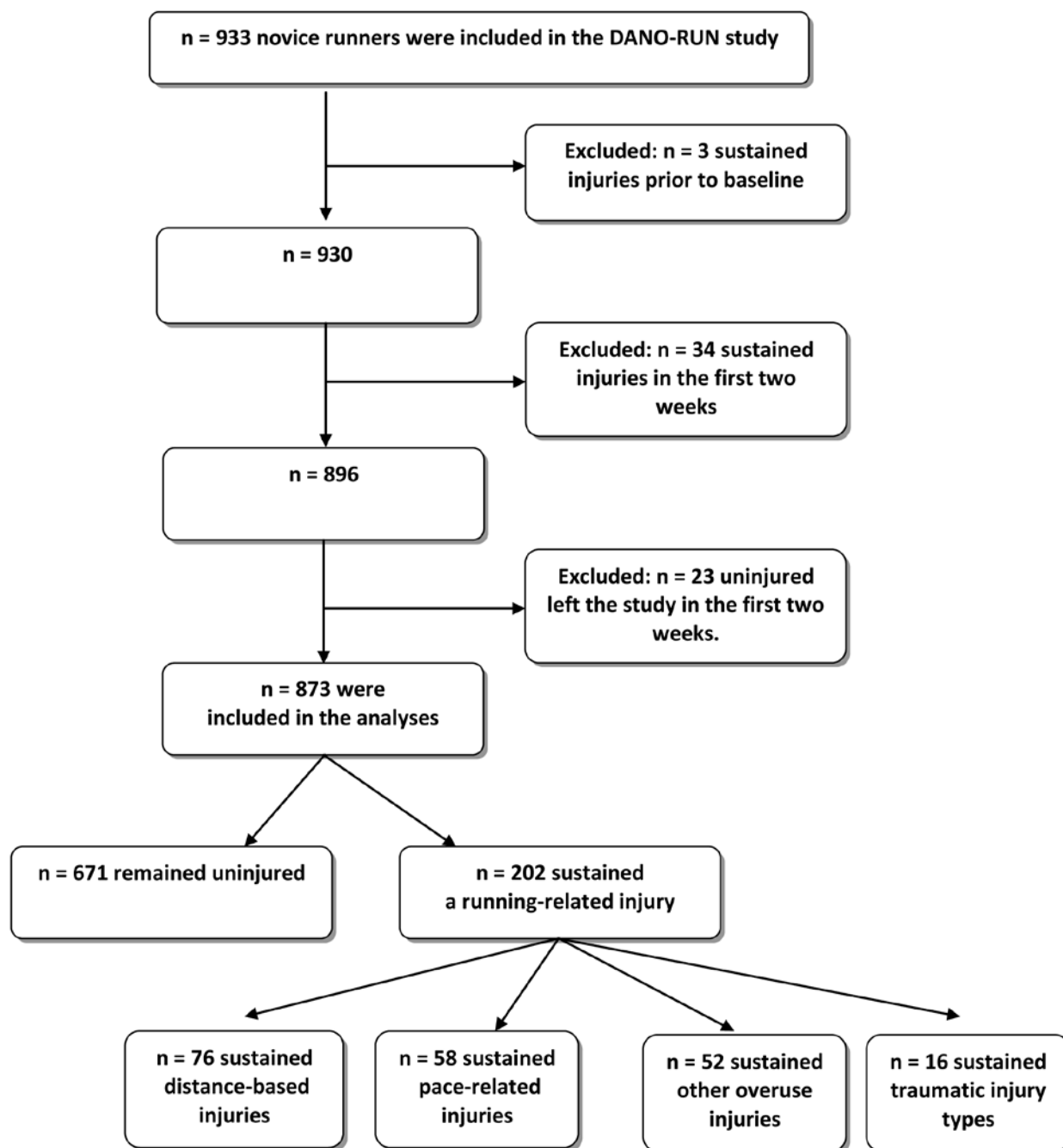


FIGURE 1: Flow-chart of the exclusions prior to analyses. When investigating the cause-specific injury propability (**TABLE 2**; analyses B, C, D, and E), the uninjured, the 76 sustaining distance-related injuries, the 58 sustaining pace-related injuries, the 52 sustaining other overuse injuries, and the 16 sustaining traumatic injuries are included.

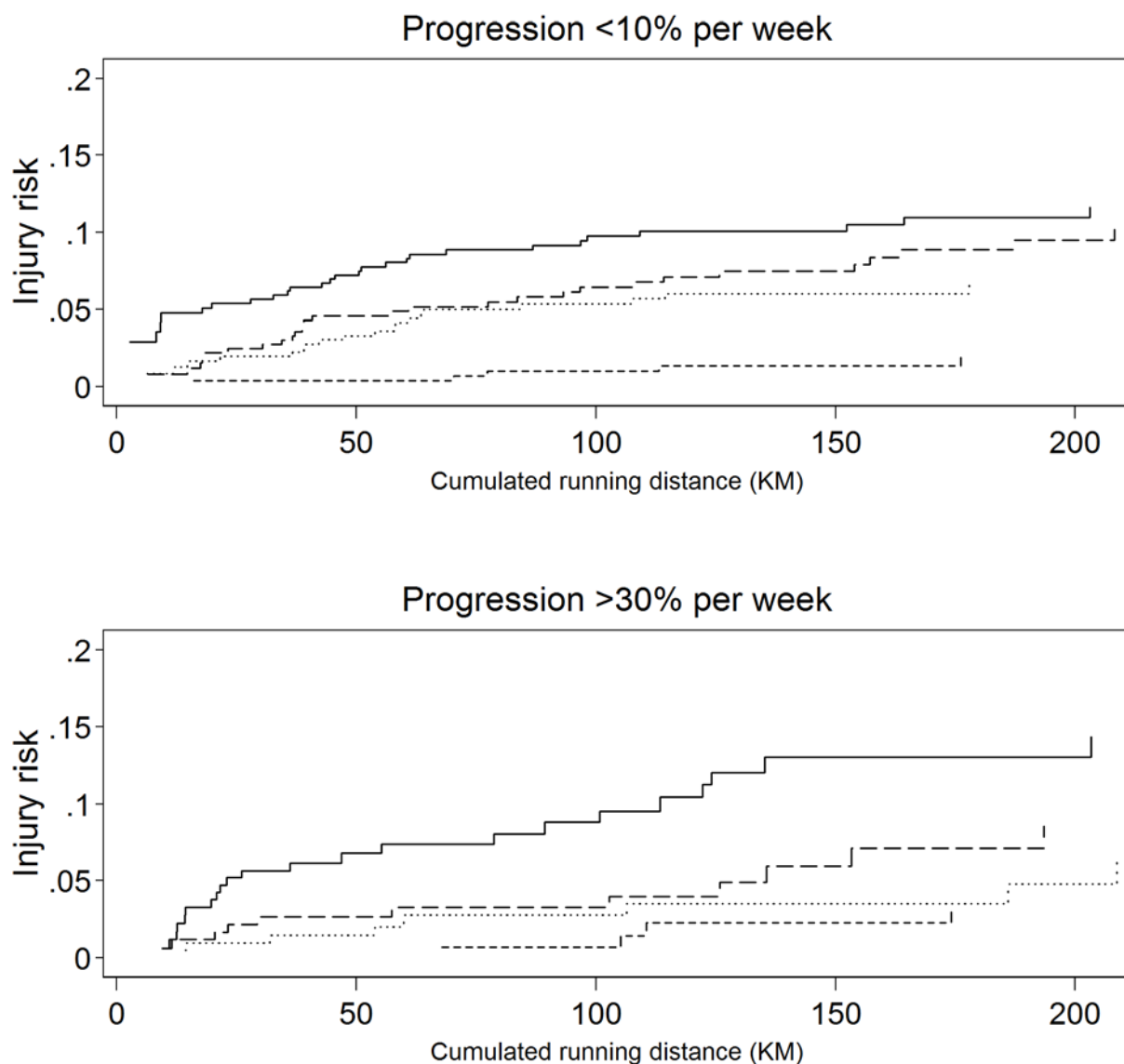


FIGURE 2: Graphs visualizing the cause-specific injury probabilities for the <10% and >30% strata during the first 200 kilometers of running.

Solid line = the cause-specific injury probability of distance-related injuries (patellofemoral pain, iliotibial band syndrome, medial tibial stress syndrome, patellar tendinopathy, gluteus medius injury, greater trochanteric bursitis, and injury to the tensor fascia latae).

Dash = the cause-specific injury probability of pace-related injuries (plantar fasciitis, Achilles tendinopathy, tibial stress fracture, hamstring injuries, iliopsoas injuries, and injuries to the triceps surae muscles).

532 Dot = the cause-specific injury probability of other types of overuse injuries.
533 Short dash = the cause-specific injury probability of traumatic injuries like ankle inversions injuries.
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